# Alvin (Haoyun) Yang

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## Research Interests

Quantum Field Theory (QFT), Quantum Gravity

## Education

**University of Illinois Urbana-Champaign (UIUC)** Urbana, IL
*Bachelor of Science in Physics* Expected: May 2027

- **Techinical GPA:** 3.91 / 4.00
- **Honors/Awards:** [Grainger College of Engineering Achievement Scholarship, FA25, FA26 Dean's List, James Scholar]

## Research Experience

**Undergraduate Research Assistant** UIUC Department of Physics
*Advisor: Prof. Tanner Trickle* [Spring 2026] – Present
*Project: Dark Matter Self Interaction (SIDM) Cross Sections*

- Currently: Independently reproduced and evaluated viscosity cross-section and scattering amplitude calculations in QFT phenomenology.
- Calculated matrix elements of different SIDM interactions using FeynCalc package.
- Verified consistency between phenomenological models and asymptotic heavy/light mediator under non-relativistic & SIDM limits.

**Future Sciencist Exchange Program (FuSEP 25) Participants** University of Science and Technology of China (USTC) Department of Modern Physics
*Advisor: Prof. Shi Pu* [July 2025]
*Project: An algebraic satisfied and gauge invariant decomposition of Angular Momentum*

- Review papers about decompositions of angular momentum, Canonical, Belinfante's and Chen et al's.
- Reproduce the idea and derivation of Chen et al's decomposition in a QED system.
- A final poster, exhibited in the final conference of the FuSEP 25 program.

## Mathematical & Theoretical Background

### Academic Baselines & Completed Courses

- **Abstract Algebra** **Grade: A+**
  *Core Mastery:* Fundamental theorem of arithmetic, congruences. Permutations. Groups and subgroups, homomorphisms. Group actions with applications. Polynomials. Rings, subrings, and ideals. Integral domains and fields. Roots of polynomials. Maximal ideals, construction of fields.
- **Complex Analysis** **Grade: A+**
  *Core Mastery:* Covers the standard topics and gives an introduction to integration by residues, the argument principle, conformal maps, and potential fields.
- **Undergraduate Core Foundations:** Classical Mechanics (A+), Quantum Mechanics (A+), Electrodynamics (A+), Statistical Mechanics (A+).

### Concurrent Advanced Enrollment (Fall 2026)

- **PHYS 515: General Relativity I (Graduate Level)**
  *Focus:* Systematic introduction to Einstein's theory, with emphasis on modern coordinate-free methods of computation. Review of special relativity, modern differential geometry, foundations of general relativity, laws of physics in the presence of a gravitational field, linearized theory, and experimental tests of gravitation theories.
- **MATH 428: Introductory Morse Theory (Honors)**
  *Focus:* Basic of Morse Theory, Influential ideas of Smale, Witten and Floer, as much geodesics of Riemannian metrics and culminate in the celebrated Periodicity Theorem of Bott as possible.
- **PHYS 508: Mathematical Physics** *(Auditing)*
  *Focus:* Core techniques of mathematical physics widely used in the physical sciences. Calculus of variations and its

applications; partial differential equations of mathematical physics (including classification and boundary conditions); separation of variables, series solutions of ordinary differential equations and Sturm-Liouville eigenproblems; Legendre polynomials, spherical harmonics, Bessel functions and their applications; normal mode eigenproblems (including the wave and diffusion equations); inhomogeneous ordinary differential equations (including variation of parameters); inhomogeneous partial differential equations and Green functions; potential theory; integral equations (including Fredholm theory).

- **PHYS 580: Quantum Mechanics (Graduate Level)** *(Auditing)*
  *Focus:* Operators, state vectors, and the formal structure of quantum theory; operator treatments of simple systems; angular momentum and vector addition coefficients; stationary state perturbation theory; introduction to scattering theory for particles without spin, partial wave analysis, and Born approximation; examples taken from atomic, nuclear, and elementary particle physics.

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[1] Last updated: September 2026